

Notes: Holmström and Tirole (QJE, 1997)

Financial Intermediation, Loanable Funds, and the Real Sector

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1 Big picture

- **Question.** How does the distribution of wealth across firms, intermediaries, and uninformed investors affect investment, financing mode, interest rates, and monitoring in an economy with moral hazard on both the borrower and lender sides?
- **Setting.** The paper builds a two-period incentive model with three agents: firms, intermediaries, and uninformed investors. Firms differ in net worth, projects require outside finance, and intermediaries can monitor firms but must themselves have “skin in the game” to monitor credibly.
- **Core challenge.** Standard corporate-finance models explain why low-net-worth firms may be credit constrained, but they do not by themselves explain why weakness in intermediary balance sheets should matter separately. Holmström and Tirole add a second agency problem: intermediaries also need capital in order to monitor.
- **Main theoretical result.** Monitoring is a partial substitute for firm collateral, but only up to the amount supported by intermediary capital. As a result, intermediary capital earns its own scarcity premium, and all forms of capital tightening raise the cutoff for obtaining finance. Poorly capitalized firms are therefore squeezed out first.
- **Why this matters.** The model cleanly distinguishes a *balance-sheet channel* (changes in firm collateral) from a *lending channel* (changes in intermediary capital). It also explains why small, collateral-poor firms are hit hardest in credit crunches.
- **Main empirical motivation.** The paper is motivated by late-1980s and early-1990s credit crunch episodes, especially in the United States and Scandinavia: bank lending fell sharply, lending was weaker where bank capital was weaker, small firms were hit disproportionately, and interest-rate spreads moved in ways consistent with tighter intermediation capacity.
- **Main comparative-statics result.** A credit crunch ($K_m \downarrow$), a collateral squeeze ($K_f \downarrow$), or a savings squeeze (an inward shift in $S(\gamma)$) all reduce investment and push the least-capitalized firms out of the market first. But the responses of the uninformed interest rate γ , the monitoring return β , and solvency ratios differ across the three shocks.
- **Policy result.** Capital adequacy ratios emerge endogenously from incentive constraints and are procyclical in the model. This suggests a rationale for why rigid solvency requirements may become especially binding in downturns.
- **Economic takeaway.** What matters is not just whether firms are poor, but also whether the financial system has enough informed capital to back monitoring. When intermediary capital is scarce, even socially good projects may go unfunded.

2 Data and measurement (key objects)

2.1 Empirical motivation rather than a new dataset

- This is mainly a theory paper. It does *not* estimate the model structurally and does *not* introduce a new micro dataset.

- Instead, it takes as motivation several stylized facts from the credit-crunch literature discussed in the introduction:
 - bank lending fell sharply in the early 1990s,
 - lending fell more where bank capital had been weaker,
 - lending shifted away from small, high-agency-cost borrowers (“flight to quality”),
 - interest-rate spread evidence was mixed but often consistent with tighter intermediation.
- The Scandinavian crises are used as a particularly vivid example because banking-sector distress, asset-price declines, and pressure on firms’ collateral all happened at once.

2.2 Key economic objects in the model

- **Firm net worth A .** This is the firm’s pledgeable internal wealth or collateral. The cross-sectional distribution is $G(A)$, and aggregate firm capital is

$$K_f = \int A dG(A).$$

- **Intermediary capital K_m .** This is informed capital. It is scarce because monitors must invest their own wealth in the projects they supervise.
- **Uninformed capital.** Outside investors supply capital either at an exogenous required return γ or through an upward-sloping supply function $S(\gamma)$.
- **Return on informed capital β .** This is the equilibrium expected rate of return on intermediary capital. It is the price of monitoring capital.
- **Financing thresholds.** The fixed-scale model generates two net-worth cutoffs:
 - $A(\gamma)$: minimum assets required for direct finance,
 - $A(\gamma, \beta)$: minimum assets required for finance with monitoring.

2.3 The three forms of capital tightening

- **Credit crunch.** A fall in intermediary capital K_m .
- **Collateral squeeze.** A fall in aggregate firm capital K_f .
- **Savings squeeze.** An inward shift in the supply of uninformed funds $S(\gamma)$.

Interpretation. These are the paper’s three distinct ways of making finance tighter. They are not observationally equivalent once one looks at interest rates, monitoring, and solvency ratios.

2.4 Why these objects are enough for the paper’s purposes

- The paper’s empirical contribution is conceptual: it maps observed lending patterns into distinct underlying balance-sheet disturbances.
- The main job of the model is therefore not to fit moments, but to derive disciplined comparative statics linking firm net worth, intermediary capital, and equilibrium credit-market outcomes.

3 Models and empirical specifications (all equations + interpretation)

3.1 Environment, agents, and timing

- There are three classes of agents: firms, intermediaries, and uninformed investors.
- There are two periods. In period 1, financial contracts are written and investment is undertaken. In period 2, returns are realized and claims are settled.
- All agents are risk neutral and protected by limited liability.
- Firms differ only in their initial assets A .

Interpretation. The two-period structure keeps the incentive problem transparent. The heterogeneity in A is the source of sorting across financing modes.

3.2 Real sector: project technology and firm moral hazard

Each firm has access to a project that costs $I > 0$ in the fixed-scale model and yields either 0 or R in period 2.

The entrepreneur can choose among three versions of the project:

- a *good* project with private benefit 0 and success probability p_H ,
- a *bad low-benefit* project with private benefit b and success probability p_L ,
- a *bad high-benefit* project with private benefit B and success probability p_L ,

with

$$\Delta p \equiv p_H - p_L > 0, \quad B > b > 0.$$

The relevant social-value assumption is

$$p_H R - \gamma I > 0 > p_L R - \gamma I + B. \tag{1}$$

Interpretation.

- Only the good project is socially efficient in the relevant range.
- Without enough incentives, the entrepreneur prefers the high-private-benefit bad project.
- The role of collateral or monitoring is to make the entrepreneur internalize enough of the upside to choose the good project.

3.3 Financial sector: monitoring technology and intermediary moral hazard

- An intermediary can monitor a firm and thereby eliminate the high-private-benefit deviation. Monitoring reduces the entrepreneur's temptation from B to b .
- Monitoring is privately costly: the intermediary incurs a nonverifiable cost $c > 0$ in order to eliminate the B -project.

- Because the intermediary's effort is itself subject to moral hazard, monitoring cannot be financed purely with other people's money. The intermediary must receive a contingent return and invest some of its own capital.

Interpretation. Monitoring relaxes the firm's agency problem, but it creates a second agency problem on the monitor side. This is the key innovation of the paper.

3.4 Direct finance

Suppose a firm borrows only from uninformed investors. In a successful state, the firm receives R_f and investors receive R_u , with

$$R_f + R_u = R,$$

and nobody is paid in failure.

The firm's incentive constraint is

$$p_H R_f \geq p_L R_f + B, \quad (2)$$

which implies

$$R_f \geq \frac{B}{\Delta p}. \quad (3)$$

So the maximum pledgeable expected income that can be promised to outside investors is

$$p_H \left(R - \frac{B}{\Delta p} \right). \quad (4)$$

Direct finance is feasible iff

$$\gamma(I - A) \leq p_H \left(R - \frac{B}{\Delta p} \right). \quad (5)$$

Define the direct-finance cutoff

$$A(\gamma) = I - \frac{p_H}{\gamma} \left(R - \frac{B}{\Delta p} \right). \quad (6)$$

Then firms with $A \geq A(\gamma)$ can invest using direct finance.

Interpretation.

- A higher required return γ raises the cutoff $A(\gamma)$.
- The firm needs enough internal wealth because some of the project's surplus must be left with the entrepreneur to preserve incentives.
- Well-capitalized firms can bypass intermediaries and borrow directly in uninformed markets.

3.5 Indirect finance: monitoring, mixed finance, and the price of informed capital

With monitoring, the successful-state payoff is split among the firm, the intermediary, and uninformed investors:

$$R_f + R_m + R_u = R.$$

The entrepreneur's incentive constraint becomes

$$R_f \geq \frac{b}{\Delta p}. \quad (7)$$

The intermediary's incentive constraint is

$$R_m \geq \frac{c}{\Delta p}. \quad (8)$$

Hence the maximum pledgeable expected income for uninformed investors is

$$p_H \left(R - \frac{b+c}{\Delta p} \right). \quad (9)$$

If the intermediary contributes I_m to the project, its expected rate of return is

$$\beta = \frac{p_H R_m}{I_m}. \quad (10)$$

Using the monitor's incentive constraint with equality gives the minimum informed investment per monitored firm:

$$I_m(\beta) = \frac{p_H c}{\Delta p \beta}. \quad (11)$$

So indirect finance is feasible iff

$$\gamma [I - A - I_m(\beta)] \leq p_H \left(R - \frac{b+c}{\Delta p} \right). \quad (12)$$

Define the indirect-finance cutoff

$$A(\gamma, \beta) = I - I_m(\beta) - \frac{p_H}{\gamma} \left(R - \frac{b+c}{\Delta p} \right). \quad (13)$$

Then firms with $A \geq A(\gamma, \beta)$ can be financed with monitoring.

The lowest acceptable return on informed capital is

$$\underline{\beta} = \frac{p_H}{p_L} \gamma > \gamma. \quad (14)$$

Interpretation.

- Monitoring lowers the entrepreneur's private gain from shirking, so it relaxes the collateral requirement.
- But monitoring uses scarce informed capital and is more expensive than uninformed finance because $\beta > \gamma$.
- Firms therefore want the *minimum* informed capital consistent with monitor incentives.
- Monitoring is socially useful only if the reduction in the entrepreneur's private benefit is large enough relative to the monitoring cost; the paper's condition is

$$c \Delta p < p_H (B - b).$$

3.6 Certification versus intermediation

The model has two equivalent interpretations of mixed finance.

- **Certification.** A sophisticated investor (for example, a venture capitalist or lead bank) puts own money into the project, certifying quality and enabling uninformed investors to provide the rest.
- **Intermediation.** Depositors place funds with a bank, which combines deposits with its own equity and then lends to monitored firms.

Interpretation. What matters is not the legal form of the contract, but the existence of informed capital with a contingent stake in project performance.

3.7 Equilibrium with fixed investment scale

The fixed-scale model sorts firms into three groups:

- **No finance:** $A < A(\gamma, \beta)$,
- **Indirect finance:** $A(\gamma, \beta) \leq A < A(\gamma)$,
- **Direct finance:** $A \geq A(\gamma)$.

Aggregate demand for informed capital is

$$D_m(\gamma, \beta) = [G(A(\gamma)) - G(A(\gamma, \beta))] I_m(\beta). \quad (15)$$

The monitoring market clears when

$$K_m = D_m(\gamma, \beta). \quad (16)$$

Aggregate demand for uninformed capital is

$$\begin{aligned} D_u(\gamma, \beta) &= \int_{A(\gamma, \beta)}^{A(\gamma)} [I - A - I_m(\beta)] dG(A) \\ &\quad + \int_{A(\gamma)}^{\infty} (I - A) dG(A), \end{aligned} \quad (17)$$

and the uninformed capital market clears when

$$D_u(\gamma, \beta) = S(\gamma). \quad (18)$$

Interpretation.

- β is pinned down by scarcity in the market for informed capital.
- γ is pinned down by the supply of uninformed funds.
- Direct and indirect finance are both equilibrium objects, and the mix depends on the distribution of firm net worth.

3.8 Comparative statics in the fixed-scale model: Proposition 1

The paper studies three forms of tightening:

- a **credit crunch** ($K_m \downarrow$),
- a **collateral squeeze** ($K_f \downarrow$),
- a **savings squeeze** (an inward shift in $S(\gamma)$).

Proposition 1. In all three cases,

- aggregate investment falls,
- the financing cutoff $A(\gamma, \beta)$ rises,
- the most poorly capitalized firms lose financing first.

Interpretation.

- This is the paper’s central “flight-to-quality” prediction.
- Small or collateral-poor firms rely most on scarce informed capital, so they are the first to be rationed out.
- The effects on interest rates are less sharp in the fixed-scale model: at least one of γ or β must rise in a squeeze, but each individual rate may move ambiguously because the distribution $G(A)$ matters.

3.9 Variable investment scale: the firm’s problem

To obtain cleaner interest-rate results, the paper next lets investment be continuous. Assume constant returns to scale:

$$R(I) = RI, \quad B(I) = BI, \quad b(I) = bI, \quad c(I) = cI.$$

A firm with assets A_0 chooses investment I , own contribution A , informed capital I_m , uninformed capital I_u , and contingent payoffs to solve

$$\max U(A_0) = p_H RI - p_H R_m - p_H R_u + \gamma(A_0 - A) \tag{19}$$

subject to

$$\begin{aligned} A &\leq A_0, \\ A + I_m + I_u &\geq I, \\ p_H R_m &\geq \beta I_m, \\ p_H R_u &\geq \gamma I_u, \\ R_m &\geq \frac{cI}{\Delta p}, \\ R_f &\geq \frac{bI}{\Delta p}, \\ R_f + R_m + R_u &\leq RI. \end{aligned}$$

All constraints bind in equilibrium, yielding the maximum sustainable investment level

$$I(A_0) = \frac{A_0}{A_1(\gamma, \beta)}, \quad (20)$$

where

$$A_1(\gamma, \beta) = 1 - \frac{p_H c}{\beta \Delta p} - \frac{p_H}{\gamma} \left(R - \frac{b+c}{\Delta p} \right). \quad (21)$$

The firm's equilibrium payoff is

$$U(A_0) = \frac{p_H b}{\Delta p} I(A_0), \quad (22)$$

and the net value of leverage is

$$\left[\frac{p_H b}{\Delta p A_1(\gamma, \beta)} - \gamma \right] A_0. \quad (23)$$

Interpretation.

- $A_1(\gamma, \beta)$ is the amount of own capital needed to support one unit of investment.
- Lower A_1 means higher leverage.
- Because of constant returns, all firms choose the same leverage ratio; firms with more assets simply scale up the same policy.

3.10 Aggregate equilibrium in the variable-scale model

Let K_f be aggregate firm capital, K_m aggregate intermediary capital, and K_u the supply of uninformed capital. Let total investment be

$$K = K_f + K_m + K_u.$$

The market-clearing condition for uninformed funds is

$$p_H(K_f + K_m + K_u) \left(R - \frac{b+c}{\Delta p} \right) = \gamma(K_u)K_u. \quad (24)$$

The equilibrium returns are

$$\gamma = \frac{p_H K \left(R - \frac{b+c}{\Delta p} \right)}{K_u}, \quad (25)$$

$$\beta = \frac{p_H c K}{\Delta p K_m}. \quad (26)$$

The aggregate firm solvency ratio is

$$r_f = \frac{K_f}{K}, \quad (27)$$

and the intermediary solvency ratio is

$$r_m = \frac{K_m}{K_m + K_u}. \quad (28)$$

Interpretation.

- With fixed monitoring intensity and constant returns, total investment depends on the *sum* $K_f + K_m$ together with the endogenous response of K_u .
- Relative scarcity between firm and intermediary capital shows up mainly in β , γ , and solvency ratios.

3.11 Comparative statics in the variable-scale model: Proposition 2

Proposition 2.

- **A. Credit crunch:** if K_m falls,
 - γ falls,
 - β rises,
 - r_m falls,
 - r_f rises.
- **B. Collateral squeeze:** if K_f falls,
 - γ falls,
 - β falls,
 - r_m rises,
 - r_f falls.
- **C. Savings squeeze:** if the supply function $S(\gamma)$ shifts inward,
 - γ rises,
 - β falls,
 - r_m rises,
 - r_f rises.

In all three cases, total investment K and uninformed capital K_u decline.

Interpretation.

- The extensive-margin result is common across shocks: financing contracts.
- But the *relative scarcity* story is different across shocks. A credit crunch makes informed capital scarcer relative to uninformed capital, so β rises. A collateral squeeze makes firms scarcer relative to intermediaries, so β falls.
- This is why observing spreads and solvency ratios helps distinguish underlying disturbances.

3.12 Endogenous monitoring and further extensions

The paper also considers a version in which monitoring intensity is endogenous. Let b vary continuously with monitoring cost c , so more intensive monitoring means lower entrepreneurial private benefits but higher intermediary cost.

- If informed capital becomes scarcer, the equilibrium response is to economize on monitoring intensity: firms choose higher b and lower c .
- If intermediary capital becomes more abundant relative to firm capital, more intensive monitoring becomes efficient.
- With endogenous monitoring, an extra dollar of intermediary capital can raise investment by more than an extra dollar of firm capital because it relaxes both the quantity of informed funds and the intensity of monitoring.
- In a decreasing-returns variant, the distribution of assets across firms matters for aggregate investment, not just the totals K_f and K_m .

Interpretation. This extension helps the authors connect the theory to episodes in which banks respond to weak collateral by shifting toward more information-intensive lending technologies.

4 Identification and instruments (what makes the results credible)

4.1 What the model identifies conceptually

- The model identifies how firm net worth maps into the choice among no finance, monitored finance, and direct finance.
- It identifies a distinct price for informed capital, β , which is absent in models where intermediaries are unconstrained.
- It identifies separate comparative statics for shocks to firm balance sheets, intermediary balance sheets, and outside savings.

4.2 How the paper separates the balance-sheet and lending channels

- The **balance-sheet channel** is captured by changes in firm collateral K_f .
- The **lending channel** is captured by changes in intermediary capital K_m .
- Because these shocks imply different movements in β , γ , r_f , and r_m , the model provides a clean framework for thinking about how to distinguish them empirically.

Interpretation. This is one of the paper’s main conceptual contributions. Earlier empirical work discussed these channels, but the model puts them into one equilibrium framework.

4.3 Why intermediary capital must matter in equilibrium

- Monitoring is privately costly and nonverifiable, so monitors need their own contingent stake.
- The paper assumes perfect correlation among the projects financed by a given intermediary. This is extreme, but it prevents the Diamond (1984) result that perfect diversification can eliminate the need for intermediary capital.
- Under this assumption, market-determined capital adequacy ratios arise endogenously from incentive compatibility.

4.4 What makes the exercise credible

- The paper does not rely on ad hoc credit rationing. Financing thresholds are derived directly from incentive constraints and limited liability.
- The model reproduces several stylized facts the authors care about: small firms depend more on intermediation, weak bank capital hurts lending, and different balance-sheet shocks can produce different spread movements.
- The certification and intermediation interpretations deliver the same allocation, which makes the economic logic robust to alternative institutional readings.

4.5 Main limitations

- The model is only two periods, so it does not fully capture dynamic feedback from interest rates to asset values and capital accumulation.
- Firm and intermediary capital are treated as exogenous in the comparative statics.
- Perfect correlation of intermediary portfolios is analytically convenient but unrealistic.
- The model abstracts from richer organizational differences across banks, venture capitalists, and other certifiers.
- The interpretation of “own capital” is most natural in an entrepreneurial setting; for real-world intermediaries, the managerial-foundations story is only sketched.

4.6 Relation to the literature

- Relative to classic credit-rationing models, the paper emphasizes balance-sheet constraints rather than interest-rate rationing alone.
- Relative to Hoshi–Kashyap–Scharfstein, Repullo–Suárez, and Diamond (1991), it adds capital-constrained intermediaries.
- Relative to Diamond (1984), it highlights the need for intermediary capital when monitoring is itself incentive constrained.
- Relative to Bernanke–Gertler and related macro-finance work, it gives a tractable equilibrium model connecting balance sheets, loanable funds, and real activity.

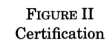
5 Tables and figures

Note. The paper has no tables. The main visual objects are Figures I–IV, and the main formal summary results are Proposition 1 and Proposition 2.

5.1 Figure I: project structure and the role of monitoring

Read:

- Interpretation.** Monitoring does not create first-best incentives on its own. It only reduces the entrepreneur's temptation, which is why collateral and monitoring are partial substitutes rather than perfect substitutes.

FIGURE I

Read:

- Interpretation.** Net worth sorts firms into financing regimes. Monitoring is most valuable for the middle group: firms too poor for market finance but not so poor that even monitoring cannot save them.

Read:

- Interpretation.** The paper's results do not depend on whether one thinks of informed capital as certification or bank intermediation. The central object is informed capital backed by own wealth.

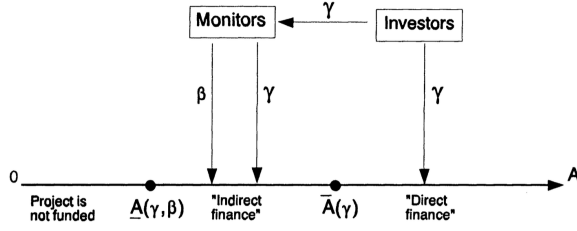


FIGURE III
Intermediation

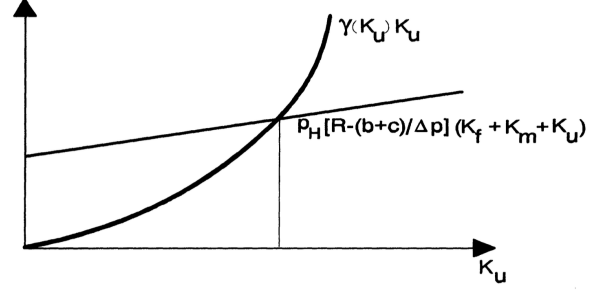


FIGURE IV

5.4 Figure IV: equilibrium in the market for uninformed capital

Read:

- Figure IV draws the supply of uninformed capital, $\gamma(K_u)K_u$, against the pledgeable-income schedule,

$$p_H \left[R - \frac{b+c}{\Delta p} \right] (K_f + K_m + K_u).$$

- Their intersection determines equilibrium uninformed capital K_u .

Interpretation. A fall in firm or intermediary capital shifts the pledgeable-income schedule and reduces the amount of uninformed capital that can be attracted. The outside market does not passively fill the gap.

5.5 Proposition 1 and Proposition 2: the two headline results

Read:

- Proposition 1 is the paper's robust extensive-margin result: every type of capital squeeze raises the financing cutoff and hits poorly capitalized firms first.
- Proposition 2 is the sharper comparative-statics result from the variable-scale model: the three capital squeezes differ in how they move γ , β , and solvency ratios.

Interpretation. Proposition 1 explains *who* gets squeezed out. Proposition 2 explains *which price and balance-sheet patterns* should accompany each kind of squeeze.

6 Short summary

- Method.** The paper builds a two-period moral-hazard model with heterogeneous firms, scarce intermediary capital, and equilibrium markets for informed and uninformed funds.
- Main conceptual result.** Monitoring is a partial substitute for collateral, but because monitors must have their own capital at risk, intermediary capital becomes an independent scarce factor of production.
- Main comparative-statics result.** Credit crunches, collateral squeezes, and savings squeezes all reduce investment and ration out the least-capitalized firms first, but they imply different movements in interest rates and solvency ratios.

- **Policy lesson.** Endogenous capital adequacy ratios are procyclical in the model, suggesting caution about imposing rigid solvency standards in recessions.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that real activity depends on the distribution of wealth across *firms*, *intermediaries*, and *uninformed investors*, not just on aggregate savings.
- It also establishes that intermediary capital is not a passive veil. When monitoring requires incentives, informed capital becomes a scarce resource that constrains lending and shapes financing patterns.
- The paper’s signature implication is that every kind of capital tightening hurts poorly capitalized firms the most, but different tightenings generate different interest-rate and solvency responses.

7.2 Economic intuition

- Firms need collateral because entrepreneurs must retain enough upside to behave diligently.
- Intermediaries can relax this requirement by monitoring, but only if *they* are given enough upside and enough own capital to monitor honestly.
- So there are really two scarce balance-sheet resources in the economy: firm net worth and intermediary net worth.
- When either one becomes scarce, financing contracts shrink, and the marginal firms to lose access are the weakest firms.

7.3 Takeaway

This paper is important because it turns “credit crunch” from a loose narrative into an equilibrium concept. Its central lesson is that loanable funds are not homogeneous: informed capital and uninformed capital play different roles, and the state of intermediary balance sheets matters in its own right. That insight is what allows Holmström and Tirole to connect firm-level financing constraints, bank lending behavior, spreads, and real investment in a single framework.